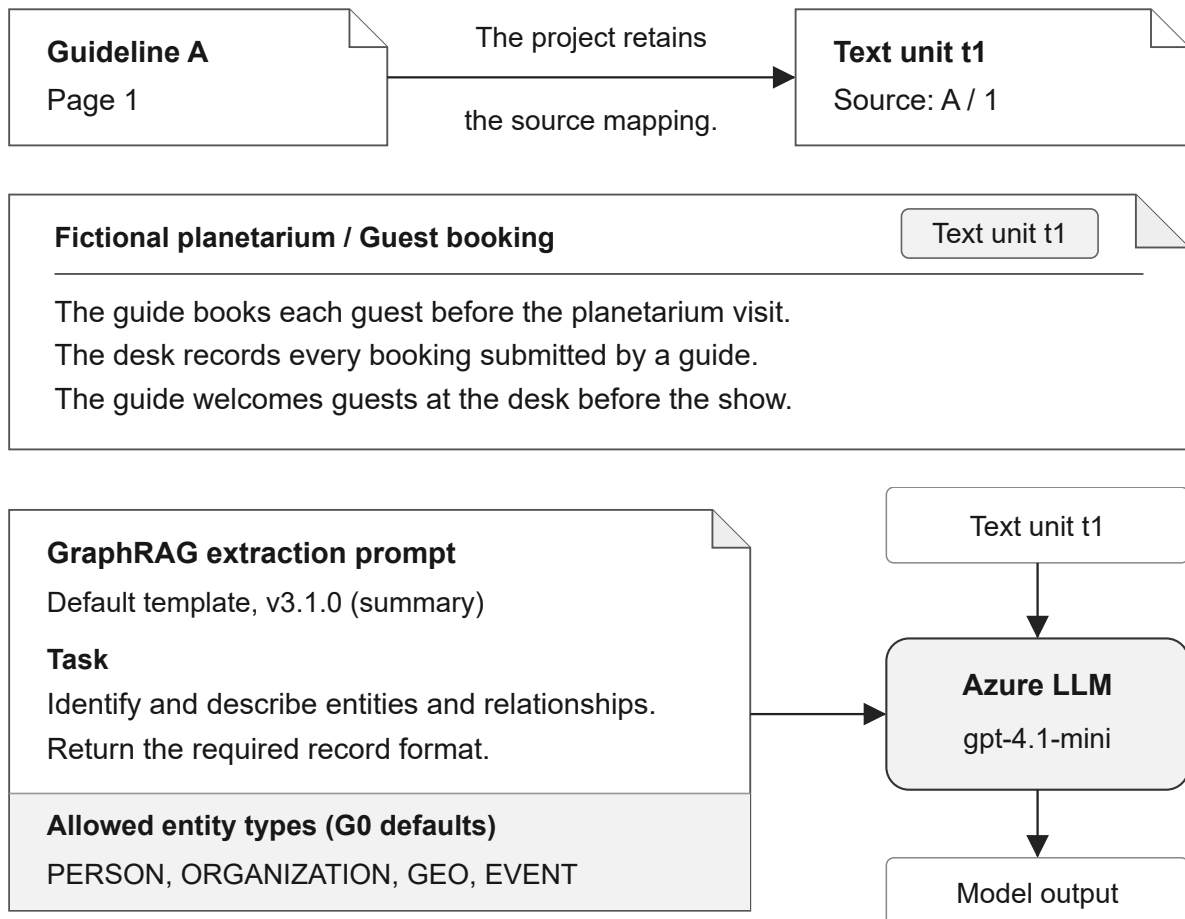


Extract records from a source text unit

Index construction

G0 uses page-based text units without overlap and retains their document-page mapping. Azure OpenAI receives the source text and native GraphRAG 3.1.0 extraction prompt [4, 35].



Illustrative model output: one entity

```
("entity"<|>GUIDE<|>PERSON<|>Person who books guests.)
```

<|> separates fields. "entity" identifies the record kind.

Figure 7. The fictional planetarium passage and structured model response are illustrative. Extraction uses the default entity types PERSON, ORGANIZATION, GEO and EVENT. The prompt panel summarises the native template.

Thesis: Section 2.5; Appendix F.

Parse and consolidate extracted records

Index construction

GraphRAG parses the response and attaches its input text-unit ID. Entity records group by name and type; relationships group by source and target names [35].

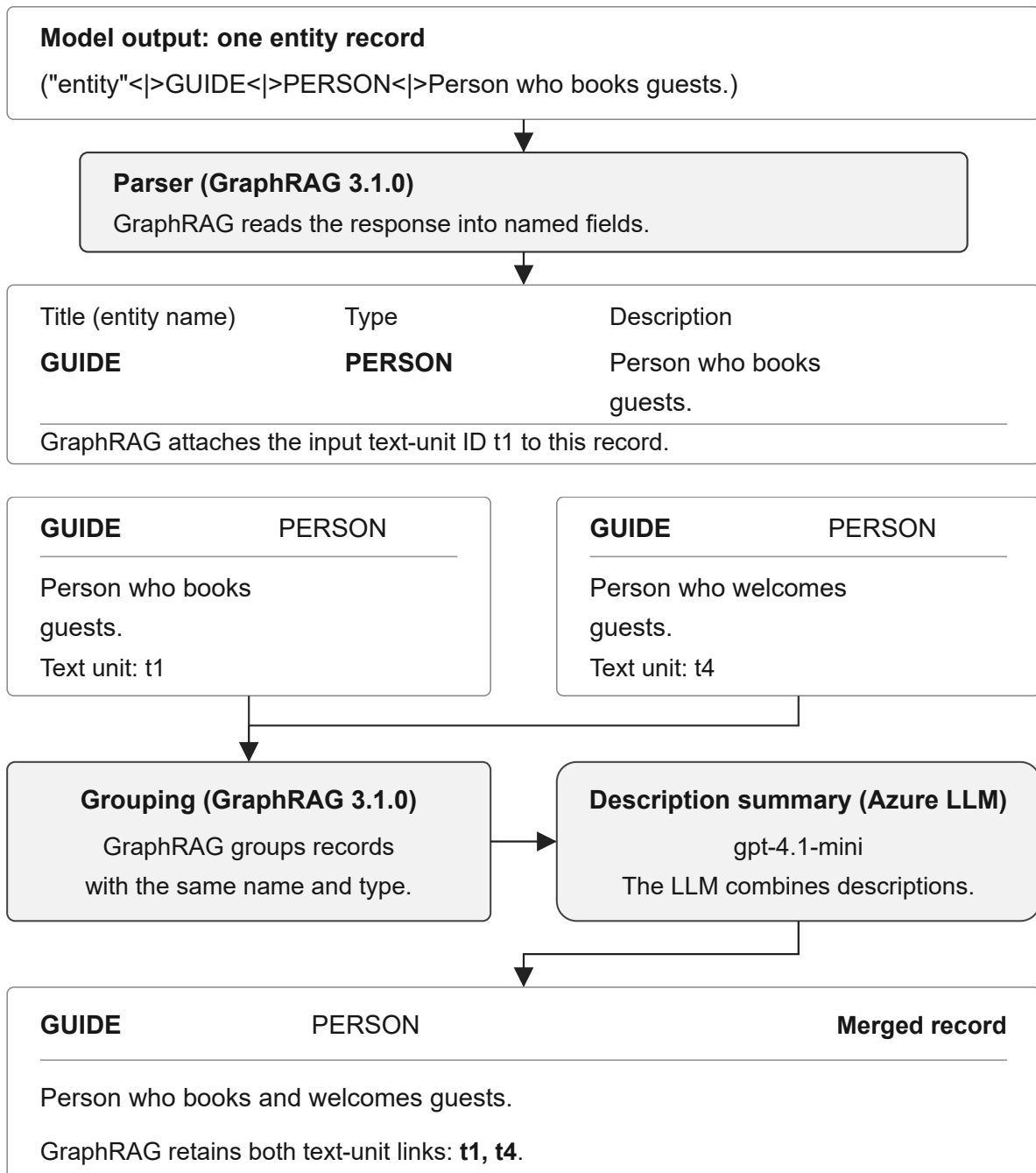


Figure 8. Identical descriptions are removed. A sole remaining description stays unchanged; differing descriptions are combined by the Azure LLM. Provenance links survive consolidation. Parsing and grouping do not verify correctness or semantic identity.

Thesis: Section 2.5; Appendix F.

Connect entities through relationships

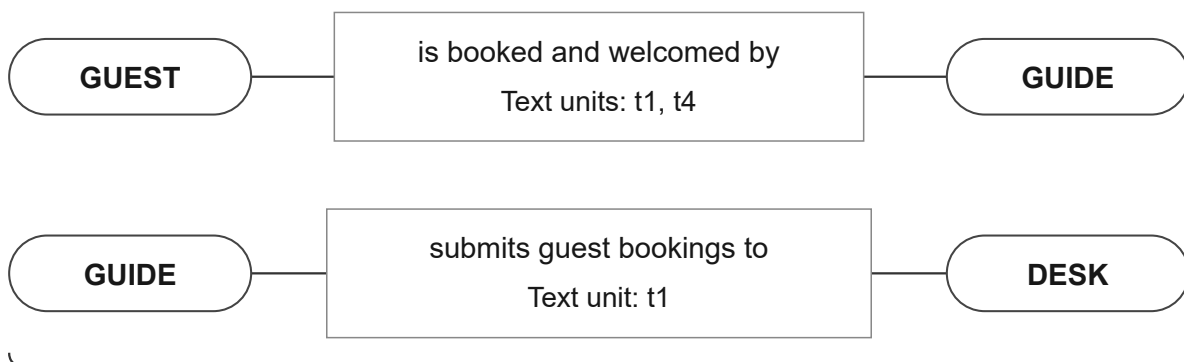
Index construction

Relationships name their two entities. Reusing the GUIDE node joins the example records into a connected graph [4, 35].

Entity records: name, type and description

GUEST Type: PERSON Person expected for a visit. Text units: t1, t4	GUIDE Type: PERSON Person who books and welcomes guests. Text units: t1, t4	DESK Type: ORGANIZATION Unit that processes guest bookings. Text unit: t1
--	---	---

Relationship records



GraphRAG uses one **GUIDE** node to join both relationships.

Result: connected graph

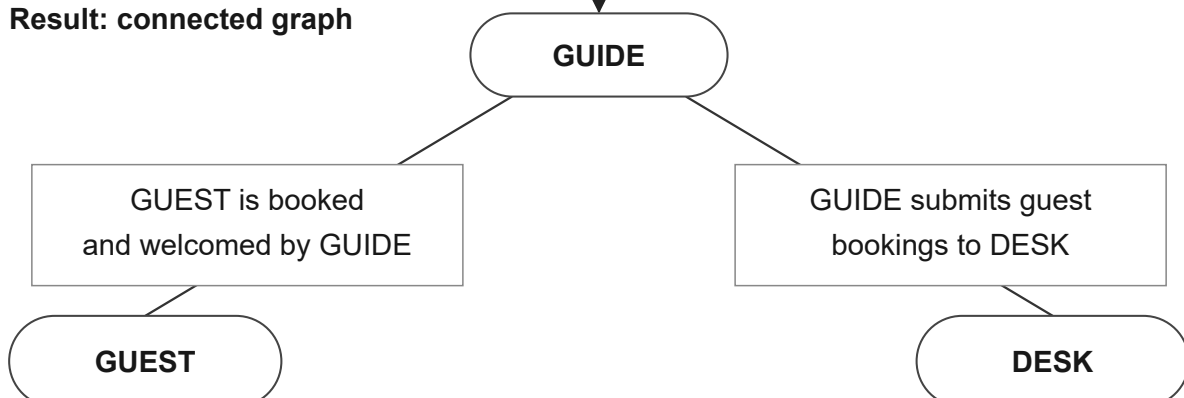


Figure 9. Edges are undirected and node positions illustrative. Descriptions and links to original text units remain attached to entity and relationship records.

Thesis: Section 2.5; Appendix F.

Detect communities and generate reports

Index construction

Hierarchical Leiden groups nodes by graph connections and relationship weights; G0 applies it to the largest connected component [36]. Azure OpenAI summarises each community [35].

Entity graph from the previous page

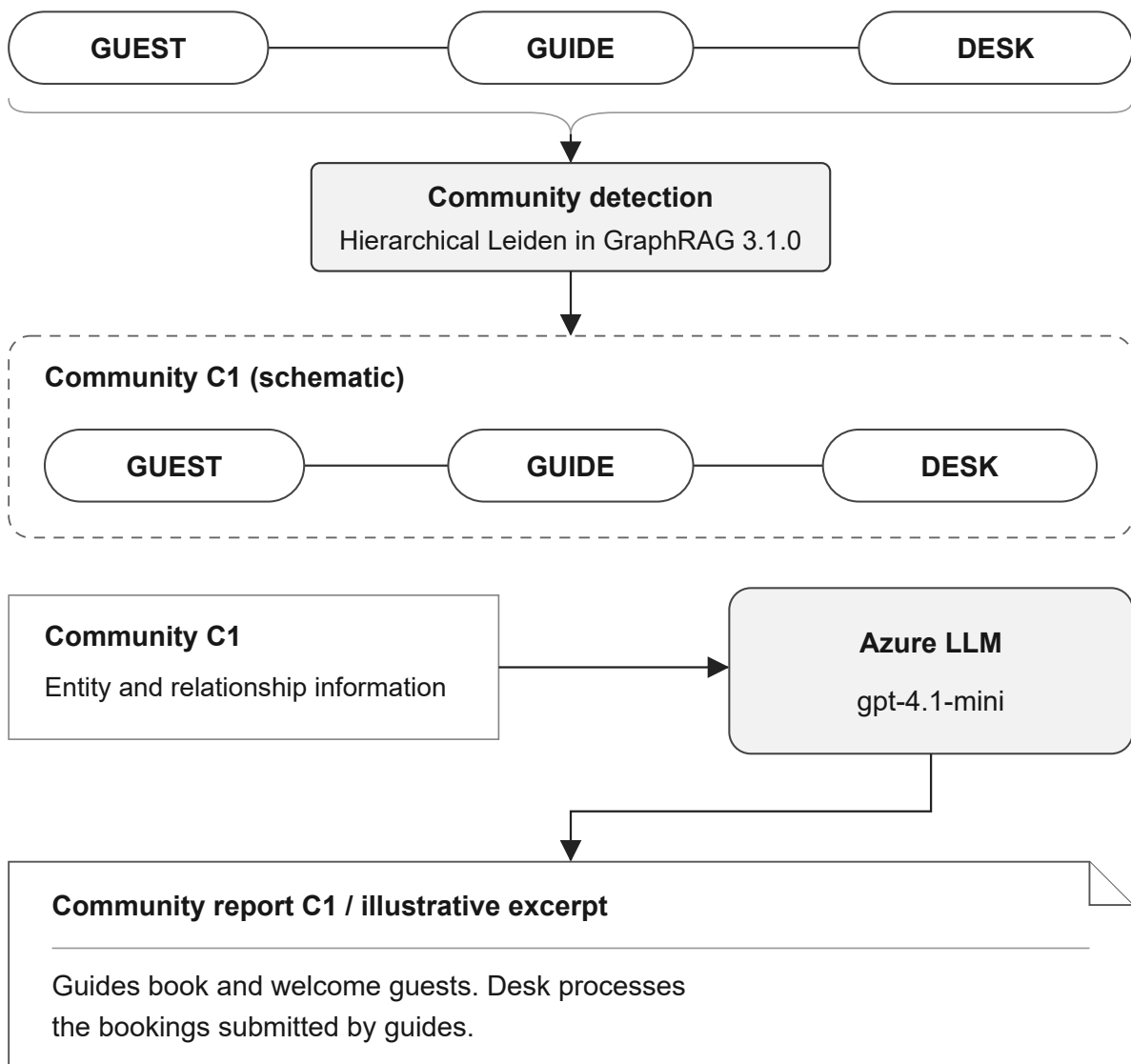


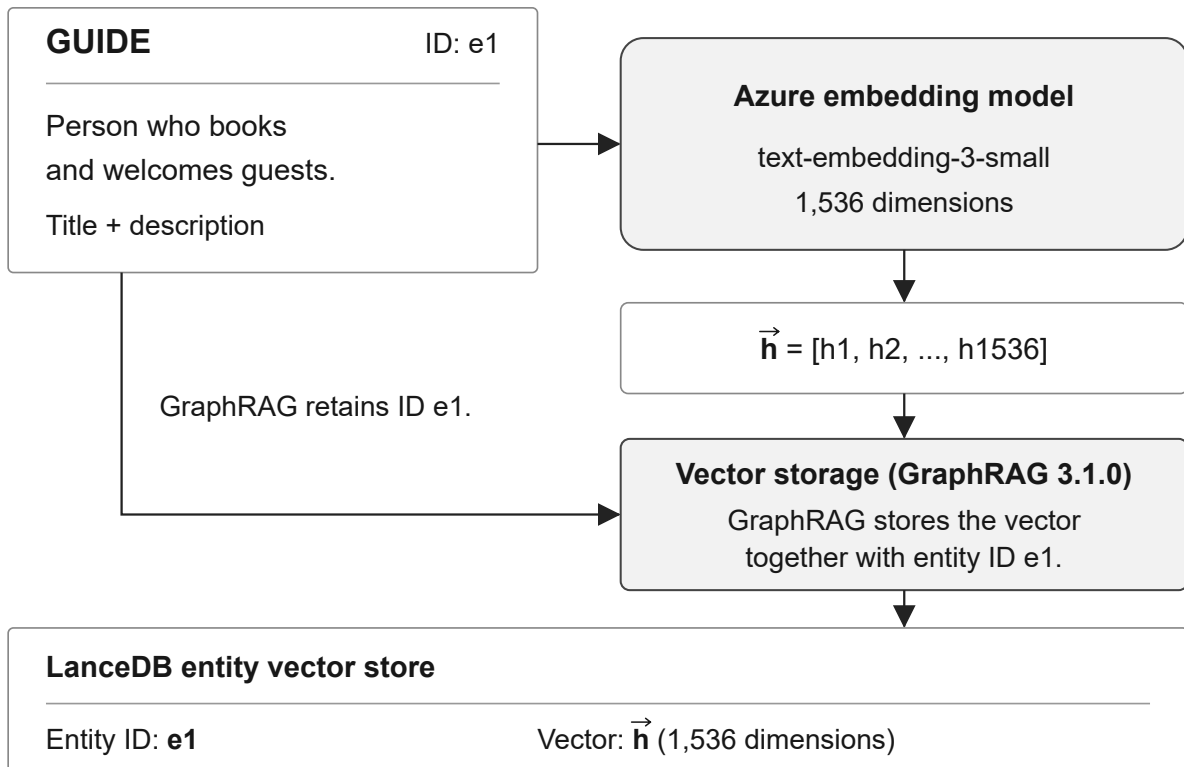
Figure 10. Only one hierarchy level is shown. C1 and its report excerpt are constructed. Membership links connect the report to its entities; reports contain derived information rather than original source text.

Thesis: Section 2.5; Appendix F.

Embed entities and retain linked outputs

Index construction

Azure OpenAI embeds each entity title and description with text-embedding-3-small. LanceDB stores the 1,536-dimensional vector with its entity ID [17, 35].



Graph records	Entities and relationships Descriptions and links to text units
Communities	Membership links and community reports For example, C1 and its report
Source records	Text units and retained source mapping t1: Guideline A, page 1 (illustrative)
Stored embeddings	Vectors for entity text, text units and reports Each vector remains linked to its record.

Result: index prepared for Local Search

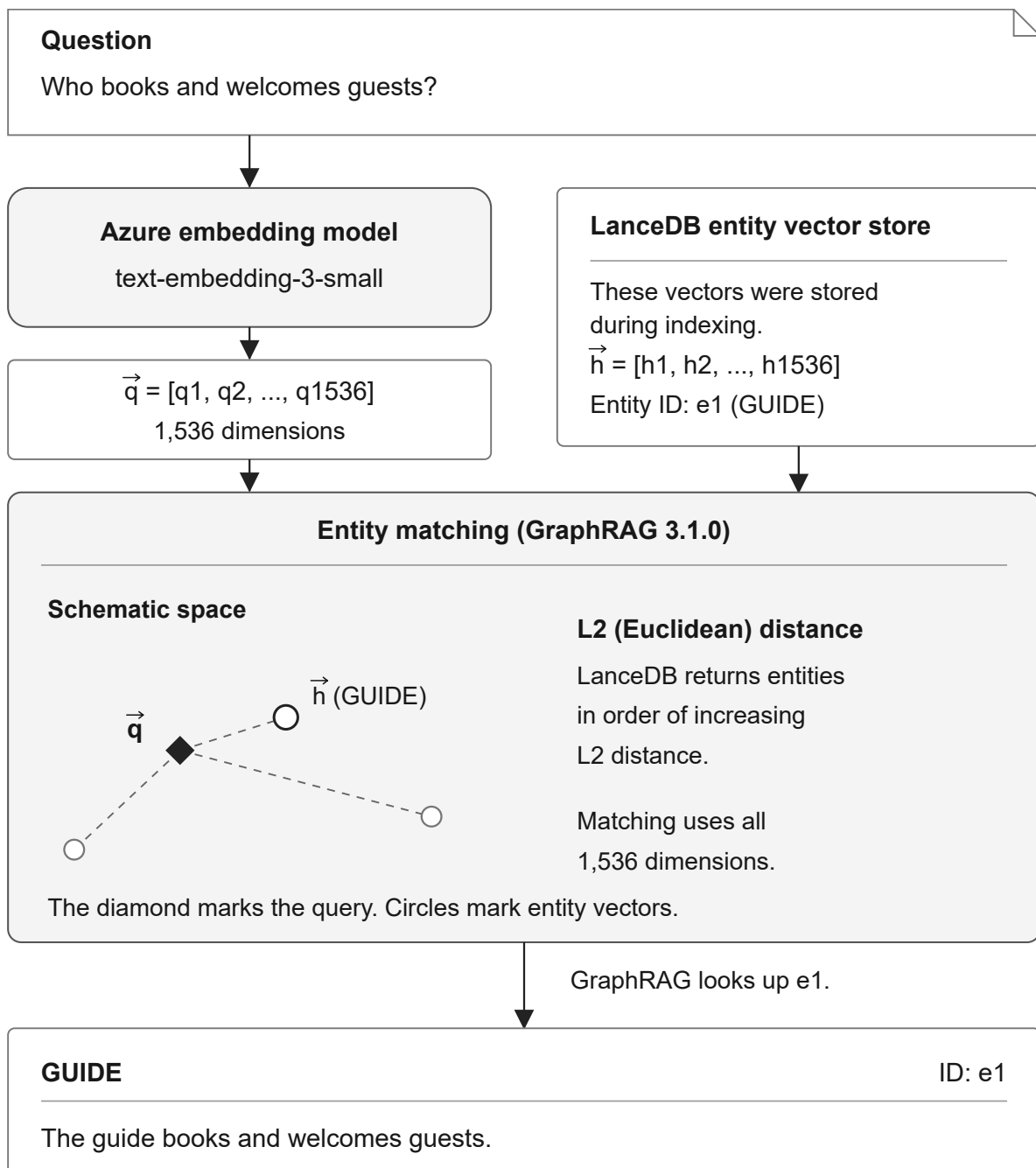
Figure 11. G0 also embeds text units and community reports. Native Local Search starts with entity-vector matching. Symbolic vectors and record IDs are illustrative. Indexing finishes before a question arrives.

Thesis: Section 2.5; Appendix F.

Find matching entities for the question

Local Search

Embed the question with the same Azure model used during indexing. LanceDB returns entity matches in increasing L2 (Euclidean) distance [35, 37].



This panel shows GUIDE as one example match.

Figure 12. The diamond marks the query; circles mark entity vectors. The spatial view is schematic and matching uses all 1,536 dimensions. GUIDE is one illustrative match, not the complete result list.

Thesis: Section 2.5; Appendix G.

Build the native mixed answer context

Local Search

Stored links connect matched entities to graph records, community reports and original text units. A token budget limits the selected native context [35].

Entry entity: GUIDE

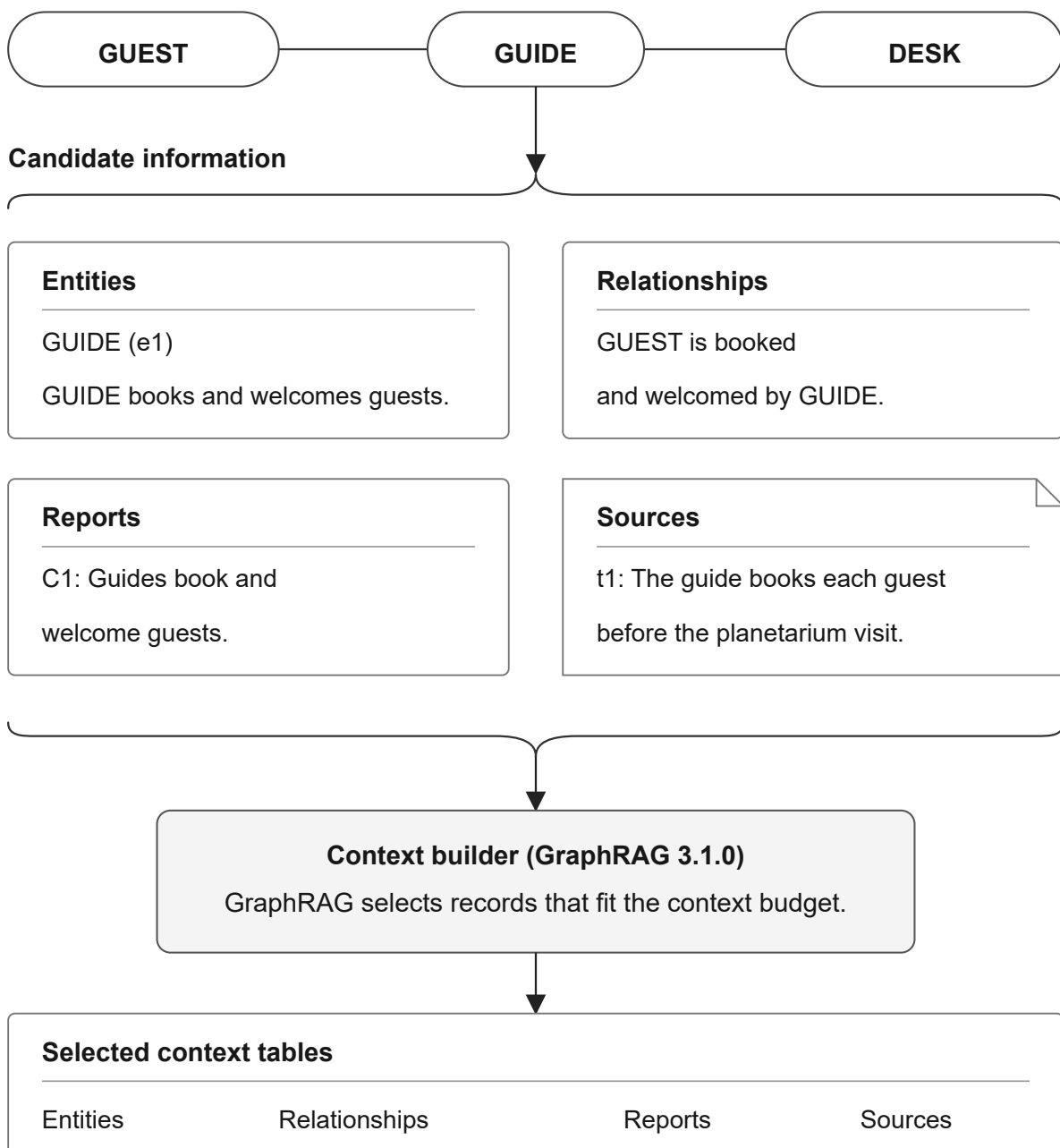


Figure 13. Sources contain original text. Entities, Relationships and Reports contain derived records. Linked candidates may be omitted to fit the budget. No evaluation projection replaces or truncates this generation context.

Thesis: Section 2.5; Appendix G.

Generate an answer with record references

Generation

Azure OpenAI receives the original question, selected context tables and native Local Search instructions [35]. The prompt panel summarises the native template.

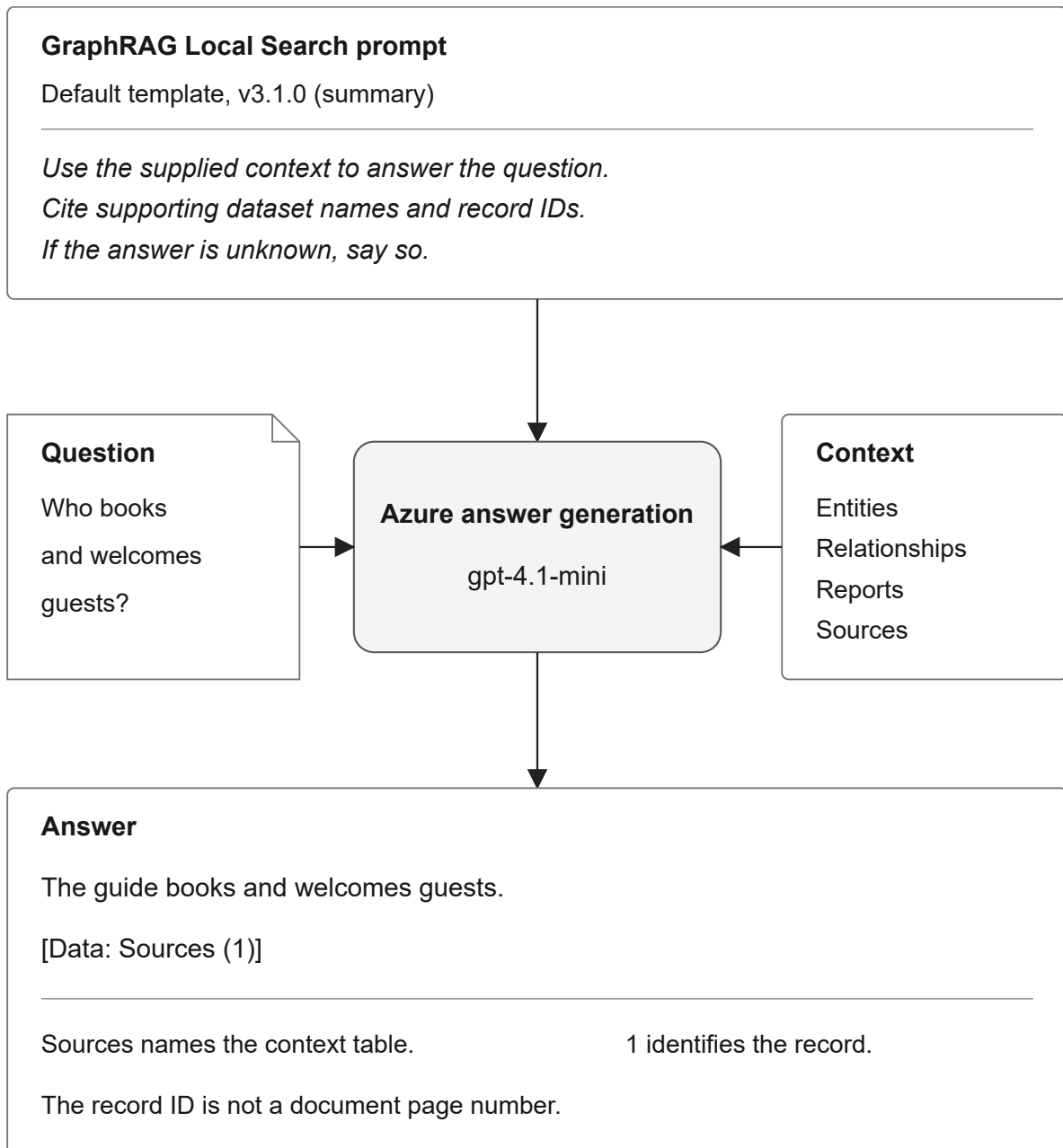


Figure 14. The constructed answer cites Sources record 1. A Sources ID identifies an original text-unit record; entity, relationship and report IDs identify derived records. Record IDs are not document page numbers.

Thesis: Section 2.5; Appendix G.

Trace a Sources record to its page

Provenance

Project provenance links connect a Sources record to its input text unit and the original document page.

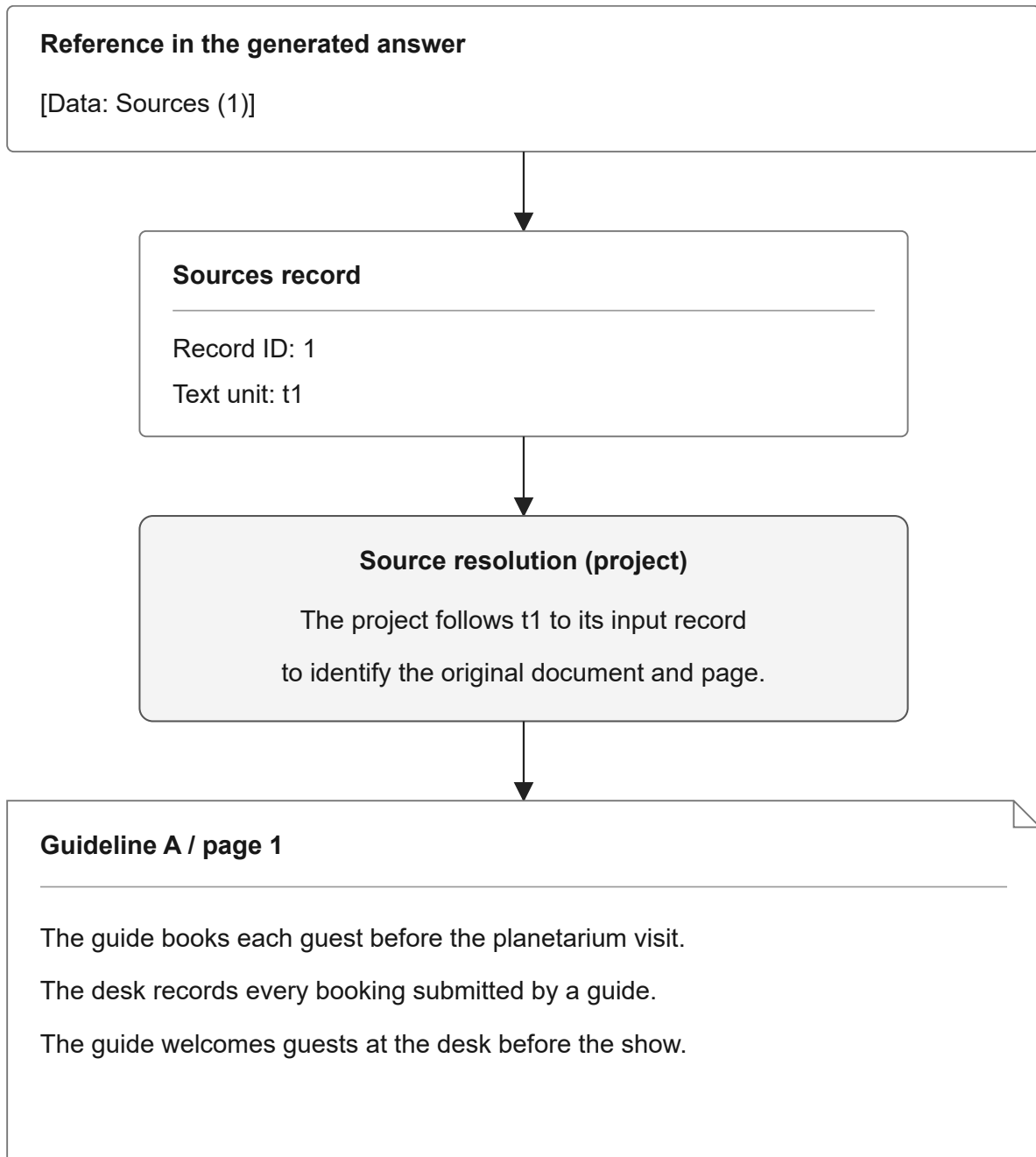


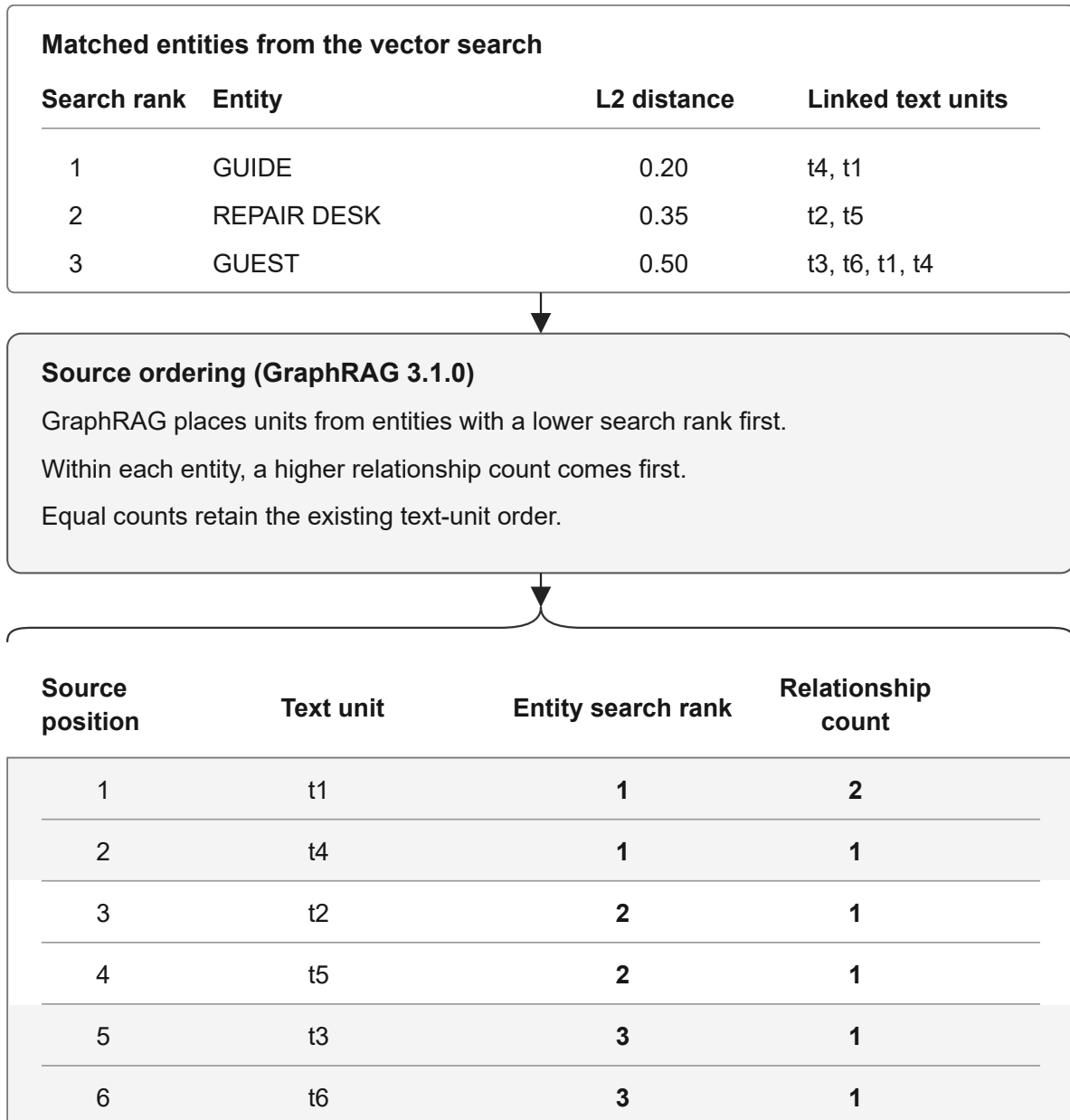
Figure 15. Resolving [Data: Sources (1)] locates the original page. Whether that page supports the answer is checked separately. All displayed passage text is fictional.

Thesis: Sections 2.4 and 2.5; Appendix G.

Establish the order of Sources

Local Search detail

Detail of Figure 13: GraphRAG assigns each text unit to its first matched entity and includes it once. It computes no separate text-unit vector score [35].



The count refers to relationships of the assigned entity linked to each unit.

For GUIDE, t1 comes before t4 because its count is 2 rather than 1.

Figure 16. Sort by ascending entity search rank, then descending relationship count for that entity and unit. Equal counts retain stored text-unit order. Append units until the next would exceed the budget. All six fit in this constructed example.

Thesis: Section 2.5; Appendix G.

Evaluate the first five Sources

Evaluation

Evaluate Sources positions 1-5 in the order established in Figure 16. Map them to shared source-page qrels without changing the native generation context.

Pos.	Source ID	Page / unit	Content (shortened)	Qrel grade
1	1	A / 1 t1	Guide books and welcomes guests.	2
2	21	D / 1 t4	Guides welcome guests at the desk.	2
3	8	B / 1 t2	Projector repairs go to the repair desk.	0
4	34	E / 3 t5	Spare bulbs are ordered via the repair desk.	0
5	14	C / 2 t3	Guest badges must show the booked session.	1
Evaluation stops after position 5.				
6	55	F / 1 t6	Guest parking is beside the planetarium.	1

nDCG@5 = 0.870

DCG / ideal DCG:
3.649 / 4.193

Evidence Recall@5 = 1

t1 supplies both facts needed
to answer the question.

Figure 17. The judged example pool contains all six pages. Linear gains give nDCG@5 = 0.870 [21]. {A / 1} covers the required facts, so Evidence Recall@5 = 1. G0 generates from all six Sources plus the other context tables; the Top-5 cutoff applies only to ranking evaluation [38]. C0 generation uses five chunks.

Thesis: Sections 2.4, 2.6 and 2.7; Appendix G.

Scope and references

Reading the figures

These figures explain the baseline workflow. Controlled variants and their configuration are documented in the repository notebooks and methods pages.

The figures use a fictional planetarium guideline. Passages, IDs, model responses, scores, counts and grades are illustrative, not measured study results. N denotes the number of indexed chunks.

Notation: folded sheets and bordered tables show documents or retained records. Rounded rectangles show processing steps. Arrows show data flow. Vector components and two-dimensional geometry are schematic.

Shared evaluation: qrel grade 2 indicates direct support for a required fact, grade 1 related content and grade 0 no answer evidence. Retrieval metrics and source traceability are separate from semantic answer assessment.

Method details: docs/classical_rag_artifacts.md, docs/graphrag_artifacts.md and docs/evaluation_methods.md. The runnable public example uses RFC 2795; it is separate from these schematic figures.

Figure and reference numbers match the thesis appendices. The full bibliography for the citations used here follows.

Master's thesis

JH. From Guidelines to Grounded Answers: Answer Quality and Configuration Effects in Vector RAG and Microsoft GraphRAG Local Search. Master's thesis, University of Applied Sciences Technikum Wien, 2026.

[4] D. Edge et al. From local to global: A Graph RAG approach to query-focused summarization. arXiv:2404.16130v2, 2025. <https://arxiv.org/abs/2404.16130>

[17] OpenAI. Vector embeddings. API documentation. <https://developers.openai.com/api/docs/guides/embeddings>

[21] K. Järvelin and J. Kekäläinen. Cumulated gain-based evaluation of IR techniques. ACM Transactions on Information Systems 20(4), 422-446, 2002. <https://doi.org/10.1145/582415.582418>

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[36] graspologic developers. Partition: hierarchical_leiden. Documentation. <https://graspologic-org.github.io/graspologic/latest/reference/reference/partition.html>

[37] LanceDB. Vector search. Documentation. <https://docs.lancedb.com/search/vector-search>

[38] C. D. Manning, P. Raghavan and H. Schütze. Introduction to Information Retrieval. Cambridge University Press, 2008, Sections 6.3 and 8.4. <https://nlp.stanford.edu/IR-book/>